Textile Border Feature Research and Implementation Architecture for Dobby and Cam Weaving Software

The translation of textile design into machine-executable instructions requires a rigorous synthesis of matrix mathematics, software architecture, and mechanical engineering. Woven fabrics are defined by the orthogonal interlacement of warp (longitudinal) and weft (transverse) threads, manipulated through a shedding mechanism that divides the warp into upper and lower layers.[1] The introduction of distinct borders—whether longitudinal selvedge borders or transverse cross borders—dramatically increases the complexity of the topological matrix.[3] To implement an advanced border design feature in computer-aided design (CAD) software that is both visually intuitive and technically accurate for real-world production, the system must algorithmically generate valid draft plans, peg plans (lifting sequences), and shaft allocations.[5] This exhaustive analysis provides the foundational logic, algorithmic pseudocode, and data modeling strategies necessary to build this architecture, ensuring strict adherence to the mechanical constraints of cam (tappet) and dobby shedding systems.

1. Typology of Borders: Weaving and Industry Perspectives

In industrial textile manufacturing, a border constitutes a distinct structural and mechanical zone within the fabric matrix. It is not merely a visual shift in color; it represents a fundamental alteration in the interlacement logic, requiring specific tension management, yarn densities, and shaft utilization.[7] Software architecture must parameterize these border types to generate valid machine instructions.

Side Borders and Selvedges

Side borders run continuously along the longitudinal edges of the woven fabric. While their primary mechanical function is to secure the structural integrity of the fabric edges to prevent unraveling during subsequent finishing processes, they are frequently utilized as primary aesthetic elements.[3]

The implementation of side borders introduces the critical challenge of differential warp take-up.[7] The take-up rate refers to the percentage of warp length consumed by the undulating path over and under the weft threads. A plain weave border features the maximum possible interlacement frequency (one over, one under), resulting in a high take-up rate. If a user designs a fabric with a plain weave side border adjacent to a twill or satin body (which possess longer floats and lower interlacement frequencies), the plain weave warp ends will rapidly become taut, while the body warp ends remain slack.[7] If woven from a single primary

warp beam, the tension differential will eventually cause the border threads to snap or the fabric to pucker severely. Software must recognize this structural divergence. Advanced CAD systems analyze the interlacement density of the respective zones and either issue a warning, recommend a secondary warp beam (supplemental warp), or automatically substitute the border structure with a weave of equivalent take-up, such as a half-basket weave.[7]

End Borders, Pallus, and Cross Borders

End borders, known traditionally in South Asian textiles as the pallu, run transversely across the width of the fabric.[9] Unlike side borders, which rely on the continuous feeding of distinct warp threads, end borders are generated entirely by altering the weft insertion sequence and fundamentally changing the lifting plan of the existing warp array.[4]

The mechanical execution of an end border requires the loom to abandon the repeating shedding logic of the body and execute a highly complex, often non-repeating sequence.[4] On a standard cam loom, this operation is generally impossible without halting the machine to physically replace the tappets.[12] On a dobby loom, the execution of a deep cross border requires an immensely long pattern chain, which historically led to the development of the multi-cylinder cross-border dobby.[12] Software systems must isolate the lifting matrix of the body from the lifting matrix of the cross border and generate the specific trigger algorithms required to switch between these data arrays seamlessly during production.[13]

Continuous vs. Intermittent Borders

The operational logic of a border can be either continuous or intermittent. Continuous borders, encompassing side stripes and standard selvedges, require a static, looping sequence within the shedding motion that operates perpetually alongside the body weave.[15] Intermittent borders, conversely, appear at calculated physical intervals, such as a dobby-woven geometric band situated precisely two inches from the hem of a terry towel.[3]

To model intermittent borders, the software must execute a spatial-to-pick translation. The system calculates the exact number of background weave repeats necessary to achieve the desired physical distance before invoking the border lifting sequence.[15] This requires the software data model to incorporate the pick density (picks per inch or picks per centimeter) to convert physical measurements into precise array indices for the lifting instructions.

Extra Warp vs. Extra Weft Borders

Highly decorative borders frequently utilize supplementary figuring threads that are not strictly necessary for the base structural integrity of the cloth.

Characteristic	Extra Warp Borders	Extra Weft Borders

Structural Basis	Embellished via an additional set of longitudinal warp threads.[17]	Embellished via an additional set of transverse weft threads.[17]
Beam Requirements	Requires two or more warp beams due to vastly different tension parameters.[17]	Requires only a single warp beam.[17]
Loom Speed	Allows for high-speed continuous weaving, maximizing productivity.[18]	Decreases loom speed significantly due to multiple weft insertions per structural row.[17]
Drafting Complexity	Drafting plans are highly complex, requiring dedicated frontal shafts for the extra threads.[18]	Drafting plans remain relatively simple; complexity shifts entirely to the lifting plan.[18]
Visual Effect	Produces prominent striped or spot patterns running the length of the fabric.[17]	Produces cross-over patterns or isolated motifs (e.g., Jamdani) running the width of the fabric.[18]

For extra warp borders, the software must draft the supplementary threads onto separate heald shafts. Because these threads often float extensively on the back of the fabric, they are typically assigned to the shafts closest to the reed to minimize the shed angle and reduce mechanical strain.[19] For extra weft borders, the software must generate lifting plans that insert ground picks and figuring picks in strict synchronization, ensuring the structural weave locks the supplementary yarn in place.

Contrast Weave Borders

A contrast weave border utilizes the exact same warp and weft yarns as the fabric body but applies an entirely different interlacement logic, such as a satin body flanked by a honeycomb border.[5] The primary limitation here is overall shaft allocation. The software must dynamically calculate the union of unique interlacement sequences required by both zones. If the body requires a minimum of 8 independent shafts and the border requires 8 independent shafts, a 16-shaft dobby is the baseline requirement.[22]

2. Draft Plan Logic with Borders

The draft plan, or threading sequence, is the foundational matrix that dictates the order in which individual warp ends are drawn through the heddle eyes of specific heald shafts.[5] When

a textile is designed with multiple zones (e.g., Left Border, Body, Right Border), the software must logically partition the total warp array and seamlessly merge multiple localized drafts into a single global draft capable of execution.

Mathematical Zoning of the Warp

In computational modeling, the draft is represented as a one-dimensional array mapping each warp end to a specific shaft index.[24] Let E represent the total number of warp ends in the fabric, and S represent the total number of available heald shafts. The global draft is an array D of length E, where $D[i] = s$ indicates that the i-th warp end is threaded through shaft s.

When borders are applied, the array E is subdivided:

- E_{lb}: Left Border ends (spanning indices 0 to $i - 1$)
- E_{body}: Body ends (spanning indices i to $j - 1$)
- E_{rb}: Right Border ends (spanning indices j to $E - 1$)

Independent vs. Amalgamated Drafting

The software engine must algorithmically determine whether the border zones and the body zone can share shafts, or if they require independent, isolated allocation.

Independent drafting is the most mechanically robust approach. If the body consists of a 4-shaft twill and the borders are an 8-shaft motif, the system allocates shafts 1 through 4 exclusively for the body, and shafts 5 through 12 exclusively for the borders.[21] This prevents the actuation of a border motif from inadvertently distorting the body structure. The global draft array concatenates the local patterns, applying a mathematical offset to the shaft indices of the border zone.

Alternatively, to conserve limited shaft resources, the software can employ amalgamated or blended drafting algorithms.[26] If a warp thread within the border zone shares the exact same binary up/down lifting sequence as a warp thread in the body zone across the entire synchronized design repeat, both threads can be assigned to the exact same shaft.[26] This represents a matrix column-matching operation. Real-world implementation of amalgamation drafting requires the software to disperse heddles across a satin base order to prevent physical crowding and friction on the loom when the thread density is exceptionally high.[29]

Step-by-Step Example of Draft Merging

Consider a scenario where a user designs a 4-shaft straight twill body flanked by 2-shaft plain

weave side borders on a loom with a maximum of 8 shafts.

1. **Left Border (6 ends):** Plain weave necessitates 2 shafts. The local draft pattern is ``.
2. **Body (8 ends):** Twill necessitates 4 shafts. The local draft pattern is ``.
3. **Right Border (6 ends):** Plain weave. The local draft pattern is ``.

The software evaluates the lifting requirements. Plain weave requires shafts that lift in strict alternation on every pick. However, in a standard 2/2 twill lifting plan, shafts 1 and 2 do not alternate in a manner that produces plain weave. Consequently, the plain weave ends cannot be amalgamated into shafts 1 and 2 of the twill block. The software must allocate independent shafts for the borders, requiring a 6-shaft loom overall.

The software generates the global draft by applying an offset to the body:

- Left Border drafted on shafts 1, 2.
- Body drafted on shafts 3, 4, 5, 6 (Offset of +2).
- Right Border drafted on shafts 1, 2.

ASCII Representation of the Global Draft Matrix (D):

Shaft 6: | | | | | | | | |X| | | | | | | | | | | |

Shaft 5: | | | | | | | |X| | | | | | | | | | | | |

Shaft 4: | | | | | | |X| | | | | | |X| | | | | | |

Shaft 3: | | | | | |X| | | | | |X| | | | | | | | |

Shaft 2: | |X| |X| |X| | | | | | | | |X| |X| |X|

Shaft 1: |X| |X| |X| | | | | | | | |X| |X| |X| |

Ends: 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0

Zone: |<-L.Border->|<- Body Twill ->|<-R.Border->|

3. Peg Plan and Lifting Plan Complexity

The peg plan, or lifting plan, is the matrix that dictates the exact configuration of raised and lowered shafts for every successive weft insertion (pick).[5] On a traditional floor loom, this is achieved via the tie-up and treadling sequence.[30] On a dobby loom, the peg plan represents a direct, one-to-one translation of the pattern, where a physical peg in a wooden lag, or an energized solenoid in an electronic dobby, commands the mechanical lifting of the corresponding heald shaft.[11]

The Constraint of Global Shaft Control

A fundamental law of frame weaving is that a heald shaft extends across the entire width of the loom. When a shaft is lifted, every single warp thread drawn through its heddles is raised simultaneously, regardless of whether that thread resides in the left border, the central body, or

the right border.[1]

Therefore, the lifting plan for a multi-zone fabric is a complex composite matrix. The software cannot independently transmit a lifting sequence to the border and a separate sequence to the body. It must compute the exact Boolean union of all active requirements for a single physical pick. Let L_{body} represent the required binary lifting state for the body shafts at pick p, and L_{border} represent the required binary lifting state for the border shafts at pick p. The global lifting vector $L_{global}[p]$ is derived via:

$$L_{global}[p] = L_{body}[p] \cup L_{border}[p]$$

Dobby Mechanics and Cross-Border Switching Logic

In a mechanical dobby mechanism, such as the Keighley or Hattersley dobby, the pattern is encoded via a continuous chain of wooden lags.[15] Each lag represents two picks (in double-lift dobbies), and holes bored into the lag correspond to the shafts.[15] A wooden peg inserted into a hole physically engages a feeling lever, which pushes a hook onto a reciprocating draw knife, subsequently lifting the jack lever and the heald frame.[12]

When an end border (pallu) is introduced, the total pick count escalates from a short repeat (e.g., 4 picks) to thousands of picks.[4] A single, continuous chain of wooden lags would become excessively heavy and lengthy, inducing severe friction. This friction commonly results in "jack missing" faults, where a bent or worn peg fails to actuate the feeler, causing the shaft to drop and the weft to float incorrectly across the warp (a defect known as stitching).[12]

To circumvent this mechanical failure, industrial weaving employs the cross-border dobby, which incorporates two or three independent pattern cylinders.[12] Cylinder A holds the short, continuous peg plan for the body cloth. Cylinder B holds the extensive peg plan for the border.[13]

The software architecture must model this hardware configuration exactly. Instead of exporting a monolithic array of 10,000 picks, the software parses the design into:

1. **Body Array:** e.g., A 4-pick repetitive loop.
2. **Border Array:** e.g., A 1,200-pick sequence.
3. **Pilot Sequence:** An instructional array that dictates exactly when the trip cam assembly shifts the mechanical drive linkage from Cylinder A to Cylinder B.[13]

For contemporary high-speed rotary electronic dobbies (operating at 1000-1500 RPM), the mechanical lag chain is entirely replaced by electromagnetic selectors governed by microprocessors (e.g., ATmega128).[15] The software exports a digital matrix (often in XML, WIF, or proprietary formats like Staubli JC5), where a binary 1 energizes the electromagnet to pull the swing arm and lift the shaft, while a 0 releases it.[31] The software's generation algorithm

must interleave the border matrices into the exact pick indices designated by the user's overarching canvas.

Step-by-Step Peg Plan Generation

Using the previous 6-shaft draft example (Plain Weave Borders on Shafts 1,2; Twill Body on Shafts 3,4,5,6), the software generates the lifting plan by mapping the desired interlacement over the draft.

- Pick 1: Plain weave requires Shaft 1 up. Twill requires Shafts 3 and 4 up.
- Pick 2: Plain weave requires Shaft 2 up. Twill requires Shafts 4 and 5 up.
- Pick 3: Plain weave requires Shaft 1 up. Twill requires Shafts 5 and 6 up.
- Pick 4: Plain weave requires Shaft 2 up. Twill requires Shafts 6 and 3 up.

ASCII Representation of the Global Lifting Plan (L):

Pick 4: | |X|X| | |X| -> Shafts 2, 3, 6

Pick 3: |X| | |X|X| | -> Shafts 1, 5, 6

Pick 2: | |X| | |X|X| -> Shafts 2, 4, 5

Pick 1: |X| |X|X| | | -> Shafts 1, 3, 4

Shaft: 1 2 3 4 5 6

4. Repeat Synchronization and Pattern Mathematics

One of the most complex algorithmic challenges in weaving CAD architecture is the synchronization of multiple weave structures possessing divergent vertical repeat lengths.[37]

The Least Common Multiple (LCM) Algorithm

Every distinct weave structure features a finite repeat size—the exact mathematical bounds of ends and picks required before the interlacement pattern cycles.[5] A standard plain weave repeats every 2 picks. A foundational 2/2 twill repeats every 4 picks. A sophisticated satin weave might repeat every 5, 7, or 8 picks.[5]

When a user digitally juxtaposes a 4-pick twill body adjacent to a 5-pick satin side border, the global lifting plan cannot merely be 4 or 5 picks in length. If the software outputs a 4-pick lifting matrix, the satin border will be abruptly truncated, resulting in a severe structural discontinuity in the cloth.[39] To ensure continuous, flawlessly synchronized patterns across the entire width of the loom, the global repeat length must be precisely calculated as the Least Common Multiple (LCM) of all individual zone repeats.[38]

Let R_{body} denote the pick repeat length of the body weave, R_{lb} denote the repeat length of the left border weave, and R_{rb} denote the repeat length of the right border weave. The total

required rows for the synchronized global lifting plan matrix R_{global} is:

$$R_{global} = \mathrm{LCM}(R_{body}, R_{lb}, R_{rb})$$

For the juxtaposition of a 4-pick twill and a 5-pick satin, the LCM is 20. The software algorithm must tile the 4-pick twill matrix 5 times vertically, and the 5-pick satin matrix 4 times vertically, constructing a unified, seamless 20-pick global lifting plan.[38]

Algorithmic Execution and Array Allocation

The CAD software must execute this LCM computation dynamically in the background every time the user modifies a border or body weave parameter.[41] If a user introduces an elaborate extra-weft floral motif in the border that cycles over 137 picks and pairs it with a 4-pick ground weave, the LCM escalates to 548. The software must instantaneously allocate memory for a 548-pick matrix and execute the tiling procedure.

Furthermore, if the user attempts to merge structurally incompatible prime-number repeats (e.g., combining 7-pick, 11-pick, and 13-pick structures), the LCM explodes to 1,001 picks. The software must calculate this LCM proactively and cross-reference it against the hardware limits of the selected loom profile. If the resulting peg plan exceeds the memory capacity of an older electronic dobby console, the system must trigger a validation error.[39]

Structural Impact of Mismatched Repeats

Beyond computational data size, forced synchronization of mismatched repeats carries severe physical consequences during production. When patterns of highly divergent interlacement frequencies are locked into synchronization, the fabric experiences localized buckling, creasing, or ruffling.[8] For example, forcing a balanced tabby (plain weave) to synchronize with a heavily floating twill border will cause the tabby zone to become physically longer and ruffle. This occurs because the high number of perpendicular intersections in the tabby prevents the warp yarn from packing down as tightly as the adjacent floats.[8] Advanced software should incorporate a predictive tension module that evaluates the ratio of interlacements per square inch across all synchronized zones, warning the user of potential structural failures prior to weaving.[42]

5. Shaft Allocation Strategy and Optimization

The ultimate bottleneck in all frame weaving, regardless of whether it employs a cam or dobby mechanism, is the finite number of available heald shafts.[2] A remarkably complex digital design is entirely useless if it mathematically necessitates 30 shafts while the user's physical mill only operates 16-shaft dobbies. To mitigate this, the software architecture must feature an aggressive shaft optimization algorithm that analyzes the composite width of the fabric and compresses the draft plan to utilize the absolute minimum number of shafts.[22]

Bipartite Graphs and Clique Partitioning Algorithms

The computational problem of minimizing heald shafts can be precisely modeled as finding the minimum biclique cover of a bipartite graph, a process fundamentally recognized as an NP-hard computational problem.[43]

The software algorithm processes the desired binary cloth simulation matrix W. Each column of matrix W represents the total vertical interlacement sequence for a single warp thread over the full LCM repeat.

1. The algorithm scans the entire width of the cloth matrix (traversing the left border, body, and right border).
2. It extracts each column as a unique boolean vector (e.g., ``).
3. It hashes these column vectors and aggregates structurally identical sequences into distinct sets.[28]
4. Every unique vector sequence is assigned to a single heald shaft. Every warp end that follows that exact sequence is drafted onto that specific shaft.[28]

By deploying advanced optimization algorithms, such as Boolean matrix factorization or heuristic clique partitioning, the software can drastically compress the draft.[43] For instance, a highly decorative border pattern that visually appears to demand 12 independent shafts might, upon rigorous mathematical inspection, contain several columns that are identical to the body's base twill. The algorithm identifies these overlaps and merges them, reducing the total shaft requirement from 16 to 10, thereby making the design feasible for a standard dobby loom.

Human Readability vs. Mathematical Optimization

While mathematical compression is technically superior, real-world textile production relies on human operators. Mill workers consistently prefer logical, human-readable drafts over perfectly compressed mathematical drafts. A highly optimized draft frequently results in a "scattered" threading sequence, where a single continuous border motif jumps erratically between shafts 1, 14, 3, and 9.[47] This disjointed sequence makes physical threading through the heddle eyes agonizingly slow and highly susceptible to catastrophic drawing-in errors.[47]

Consequently, the software architecture must support a dual-mode allocation strategy [21]:

- **Maximum Compression Mode:** Employs graph theory to condense the design into the absolute minimum shafts, prioritizing machine execution over human readability.[43]
- **Block/Zone Mode:** Overrides maximum compression to allocate continuous, dedicated blocks of shafts to specific zones. The algorithm isolates the body (e.g., forcing it onto shafts 1-4) and isolates the borders (e.g., forcing them onto shafts 5-12). While this uses more shafts, it ensures the draft is physically intuitive and rapidly executable by the mill staff.[21]

6. Hardware Constraints: Dobby vs. Cam Systems

A defining hallmark of professional textile CAD software is its capability to constrain the digital canvas based on the unyielding physical realities of the target hardware.[12] The software must differentiate strictly between the mechanical capabilities of a cam (tappet) loom and a dobby loom.

Parameter	Cam (Tappet) Mechanism	Dobby Mechanism
Maximum Shafts	Strictly limited; generally supports a maximum of only 8 heald shafts.[12]	Highly versatile; typically controls 12 to 24 shafts, with specialized units supporting up to 40 shafts.[12]
Pick Repeat Length	Rigidly fixed; dictated by the physical circumference and gearing of the cam (typically 4, 5, 6, or 8 picks).[12]	Virtually infinite; limited only by the length of the wooden lag chain or the memory of the electronic controller.[11]
Border Feasibility	Side borders are feasible if the combined shaft count is $\leq$ 8. Cross borders are virtually impossible.[12]	Both complex side borders and deep cross borders are highly feasible.[12]
Operational Speed	Runs efficiently at very high speeds with minimal machine vibration.[12]	Older mechanical units suffer from extreme vibration and friction at high speeds.[12]
Design Alteration	Requires halting the machine, dismantling the bottom shaft, and physically replacing the steel tappets.[12]	Allows rapid modification by simply swapping the peg chain or loading a new digital file.[12]

Cam Loom Software Validations

Software targeted at cam loom production must enforce immutable mathematical validations. If the calculated unique shaft count (body + borders) exceeds 8, or if the user attempts to insert a transverse cross border requiring a change in the repeat sequence, the system must throw a

fatal validation error, preventing the export of the draft.[42] The software must understand that the physical geometry of a tappet cannot dynamically alter its lifting profile mid-production.[12]

Dobby Loom Software Validations

While dobbies offer expansive design freedom, they introduce mechanical vulnerabilities. In negative cam dobbies, the heald frame is lifted by the pattern selection but lowered by heavy reversing springs.[15] The software must calculate the load distribution across the lifting cycle. If a heavy cross-border design necessitates 20 shafts to be lifted simultaneously against spring tension for 500 consecutive picks, the software should warn the user of impending mechanical strain. Such configurations frequently lead to the "jack missing" fault or excessive vibration, degrading the quality of the shed.[12]

7. Software Architecture and Implementation

To translate these complex mechanical constraints and mathematical models into a modern web application (e.g., utilizing React on the frontend with a Node.js or Python backend), the software architecture must be rigorously decoupled. The system must segregate the visual representation (the UI canvas) from the underlying mathematical models (the matrix engine) and the hardware export modules (the machine drivers).[50]

Data Model Design

The state of the fabric should be parameterized using strict, typed JSON structures that separate the conceptual design zones from the compiled, machine-ready matrices.[51] By storing the design as objects rather than a monolithic pixel grid, the software can rapidly recalculate parameters when a single zone is altered.

JSON

```
{
 "fabricDefinition": {
  "dimensions": {
   "totalWarpEnds": 2400,
   "picksPerInch": 60
  },
  "targetHardware": {
   "loomType": "ELECTRONIC_DOBBY",
   "maxShafts": 24,
   "maxPickMemory": 5000
  },
```

```
  "zones": {
    "leftBorder": {
      "widthEnds": 200,
      "weaveMatrix": [,]
    },
    "body": {
      "widthEnds": 2000,
      "weaveMatrix": [,,,]
    },
    "rightBorder": {
      "widthEnds": 200,
      "weaveMatrix": [,]
    },
    "crossBorders": [
      { "startPickIndex": 1200, "lengthPicks": 300, "weaveMatrix": [...] }
    ]
  }
 }
}
```

The compilation engine processes this conceptual state into the required mechanical format, generating the draft and peg plan objects:

JSON

```
{
 "compiledMachineOutput": {
   "totalShaftsUtilized": 16,
   "draftArray": [0, 1, 0, 1, 2, 3, 4, 5, 2, 3, 4, 5...],
   "totalPicksGenerated": 1500,
   "liftingMatrix": [1, 0, 1, 0, 0, 0, 1, 1...],
    [0, 1, 0, 1, 1, 1, 0, 0...]
 }
}
```

Algorithm Design: The Compilation Engine

The core algorithm compiling the zones into the final mechanical output executes in three distinct phases: Matrix Expansion, Shaft Optimization, and Lift Plan Generation.[51]

Phase 1: Matrix Expansion (Synchronizing Repeats) The software extracts the small

repeating weave matrices of the Left Border, Body, and Right Border. It calculates the LCM of the pick repeats to determine the vertical height, and uses the user-defined warp end counts to determine the horizontal width.[38] It then executes modulo arithmetic to tile the arrays across the global matrix.[41]

Pseudocode for Matrix Expansion:

TypeScript

```
function expandFabricMatrix(zones, lcmRepeat) {
  let globalMatrix =;
  for (let row = 0; row < lcmRepeat; row++) {
    let currentRow =;

    // Tile Left Border Sequence
    for (let col = 0; col < zones.leftBorder.widthEnds; col++) {
      let localRow = row % zones.leftBorder.weaveMatrix.length;
      let localCol = col % zones.leftBorder.weaveMatrix.length;
      currentRow.push(zones.leftBorder.weaveMatrix[localCol]);
    }

    // Tile Body Sequence
    for (let col = 0; col < zones.body.widthEnds; col++) {
      let localRow = row % zones.body.weaveMatrix.length;
      let localCol = col % zones.body.weaveMatrix.length;
      currentRow.push(zones.body.weaveMatrix[localCol]);
    }

    // Tile Right Border Sequence (similar modulo tiling)
    //...

    globalMatrix.push(currentRow);
  }
  return globalMatrix;
}
```

Phase 2: Shaft Allocation and Draft Generation The algorithm transposes the globalMatrix so that the warp ends (columns) become iterable rows. It employs a hashing function to compress identical intersection sequences, mapping each unique sequence to a sequential shaft index.[28] If the required shafts exceed targetHardware.maxShafts, the process terminates and throws an

error to the UI.

Pseudocode for Shaft Allocation:

TypeScript

```
function generateOptimizedDraft(globalMatrix, maxShafts) {
  let draftArray =;
  let uniqueShaftMap = new Map();
  let shaftCounter = 0;

  let transposedMatrix = transposeMatrix(globalMatrix);

  for (let end = 0; end < transposedMatrix.length; end++) {
    let sequenceHash = hashBooleanArray(transposedMatrix[end]);

    if (!uniqueShaftMap.has(sequenceHash)) {
      if (shaftCounter >= maxShafts) {
        throw new Error("Hardware Error: Maximum shaft limit exceeded.");
      }
      uniqueShaftMap.set(sequenceHash, shaftCounter);
      shaftCounter++;
    }
    draftArray.push(uniqueShaftMap.get(sequenceHash));
  }
  return { draftArray, totalShafts: shaftCounter, shaftDefinitions: uniqueShaftMap };
}
```

Phase 3: Lifting Plan Generation The algorithm iterates through the original globalMatrix row by row. For each pick, it evaluates the boolean state of every warp end. If an end must be raised (1), it cross-references the draftArray to determine which shaft houses that end, and subsequently marks that shaft as 1 in the final liftingMatrix for that specific pick.[33]

UI/UX Interface Architecture

The user interface must obscure this intense matrix mathematics behind an intuitive, visual workspace.[55]

- **Visual Zoning:** The workspace should render a primary grid representing the fabric body, flanked by draggable vertical and horizontal bounds. Users drag these boundaries to define the physical width of the side borders and the height of the pallu/cross borders in real-time.

- **Drag-and-Drop Library:** A component panel allows users to drag pre-defined weave structures (e.g., a 2/2 twill, a 5-end satin) and drop them directly into specific zones.[56] The underlying engine instantaneously recalculates the LCM, re-runs the hashing algorithm, and updates the visualization.[58]
- **Real-time Validation Badges:** As the user designs, a dynamic badge tracks hardware limits: Shafts Utilized: 18/24. If the user drops a highly complex floral motif into the border that spikes the requirement to 28 shafts, the border zone highlights in red, the badge flashes a warning, and the compilation engine halts, preventing the generation of an un-weavable state.[57]
- **3D Yarn Simulation:** Leveraging WebGL or Three.js, the 2D interlacement matrix is extruded into a 3D simulation reflecting actual yarn thickness and color.[59] This enables the user to visually audit the fabric for dangerous, unsupported long floats that require structural binding points.

8. Advanced Features and Output Generation

To elevate the software from a mere drafting utility to an advanced industrial CAD system (akin to platforms like ArahWeave, ScotWeave, or DigiBunai), several predictive features must be integrated.[6]

Algorithmic Border Generation

Instead of requiring the user to map border pixels manually, the software should include parameterized procedural generators.[56] Utilizing algorithms similar to Cellular Automata, the software can automatically generate complex, scalable geometric borders (such as Greek Keys, temples, or chevrons).[40] The user inputs the desired motif style and target width, and the software mathematically outputs the required matrix, instantly validating it against the available shaft constraints.

The Weft Blanket Tool and Continuous Sampling

Industrial sampling is an expensive process. Advanced software implements a "Weft Blanket" feature.[57] This algorithm allows the user to design multiple variations of cross borders (e.g., trying five different floral motifs) and sequentially compiles them into one massive, continuous digital file. This enables the loom to weave all five variations sequentially without stopping the machine, drastically reducing sampling overhead.[57]

Output Formatting and Machine Drivers

The final architectural requirement is translating the compiled JSON matrices into industrial machine code. The software must transpose the binary liftingMatrix into the precise byte-code formats demanded by electronic dobby controllers (e.g., standard WIF format, Staubli JC5, or specific serial formats for ATmega-driven looms).[32] For manual or mechanical cross-border dobbies, the software must generate a printable ASCII layout or PDF report detailing exactly

how the mill worker should peg the primary cylinder versus the pilot cylinder to ensure accurate execution on the mill floor.[11]

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